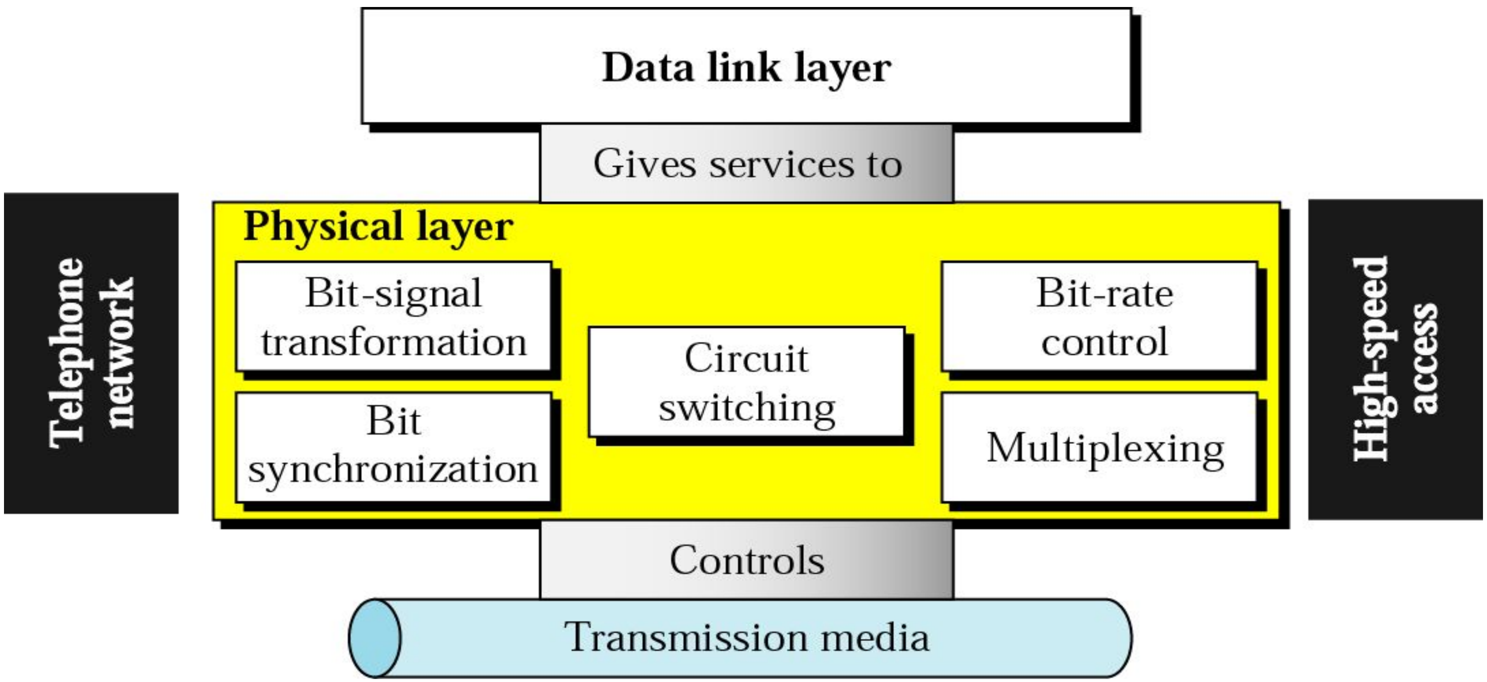
Computer Communication & Networks

Lecture 10

Physical Layer: Transmission Media

Physical Layer



Physical Layer Topics to Cover

Signals

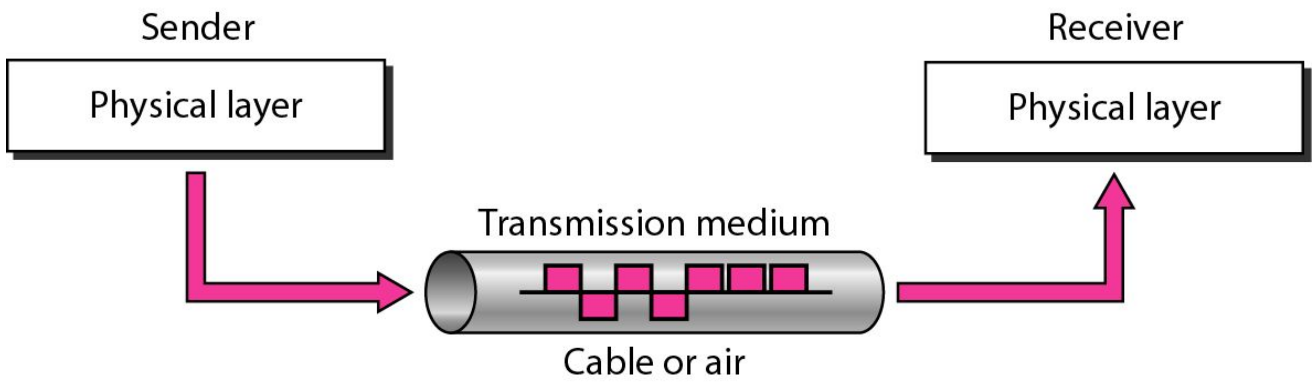
Digital Transmission

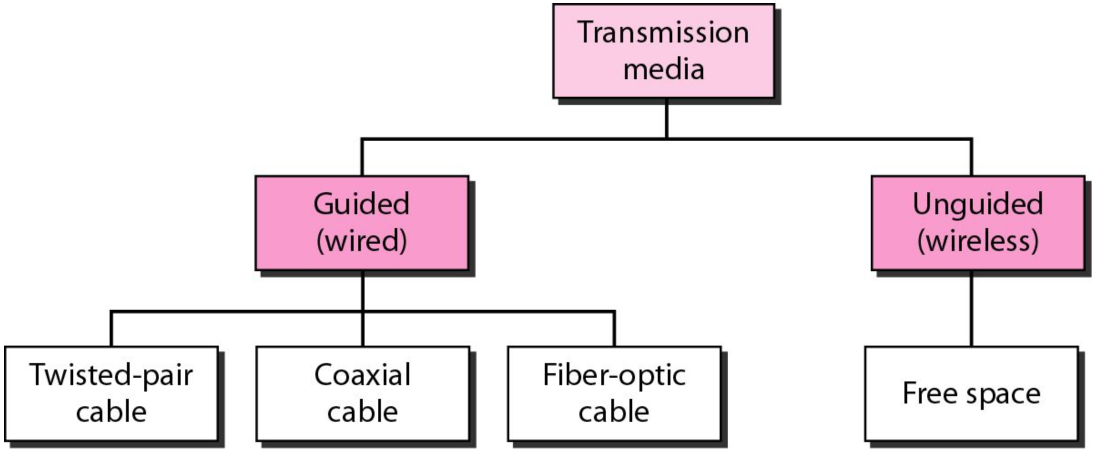
Analog Transmission

Multiplexing

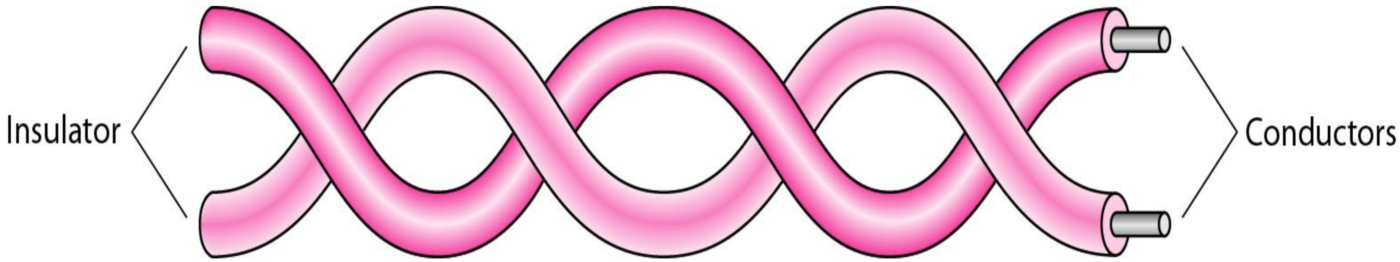
Transmission Media

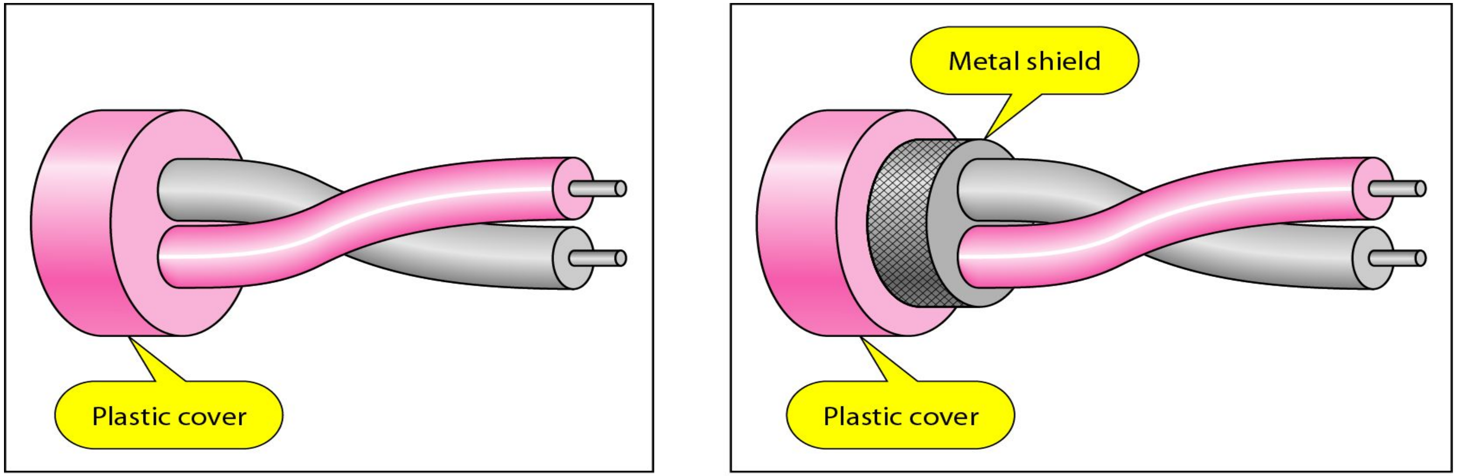
Transmission Medium and Physical Layer





Twisted-pair Cable





a. UTP b. STP

Categories of unshielded twisted-pair cables

Category	*Specification*	*Data Rate (Mbps)*	*Use*
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T-lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called SSTP (shielded screen twisted-pair). Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

Twisted Pair Cable

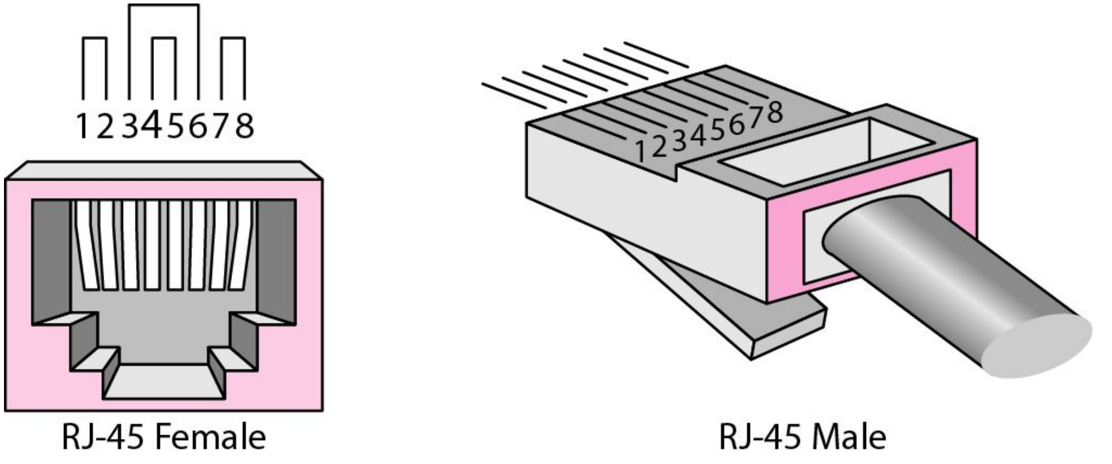
(a)

(b)

(a) Category 3 UTP

(b) Category 5 UTP

UTP connector



Twisted Pair Cables (Example)

- **ADSL**
- **Ethernet networks**

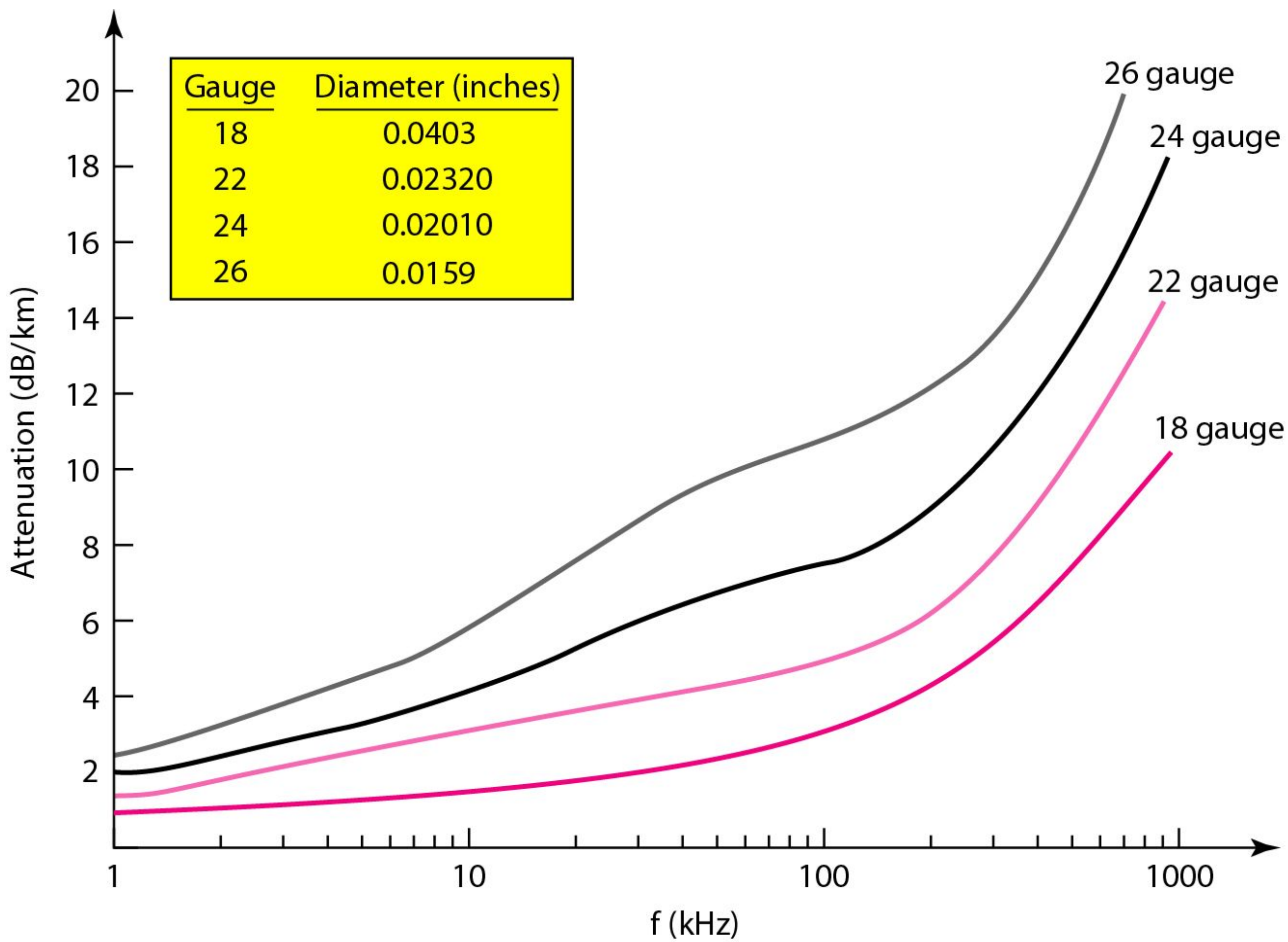
  - 10BASE-T

  - 100BASE-TX

  - 1000BASE-T

  - 1000BASE-TX (Cat5e (enhanced))

UTP Performance



Twisted Pair Cable (Pros & Cons)

Pros:

- easy to understand
- mass production - low cost
- most widely used medium

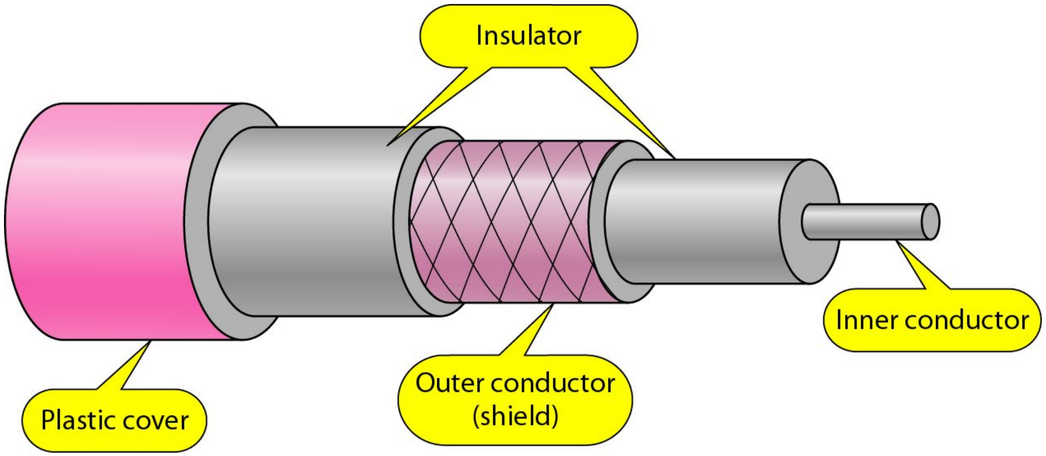
Cons:

- prone to electromagnetic interference
  - in power plants, airport buildings, military facilities, cars…

Note:

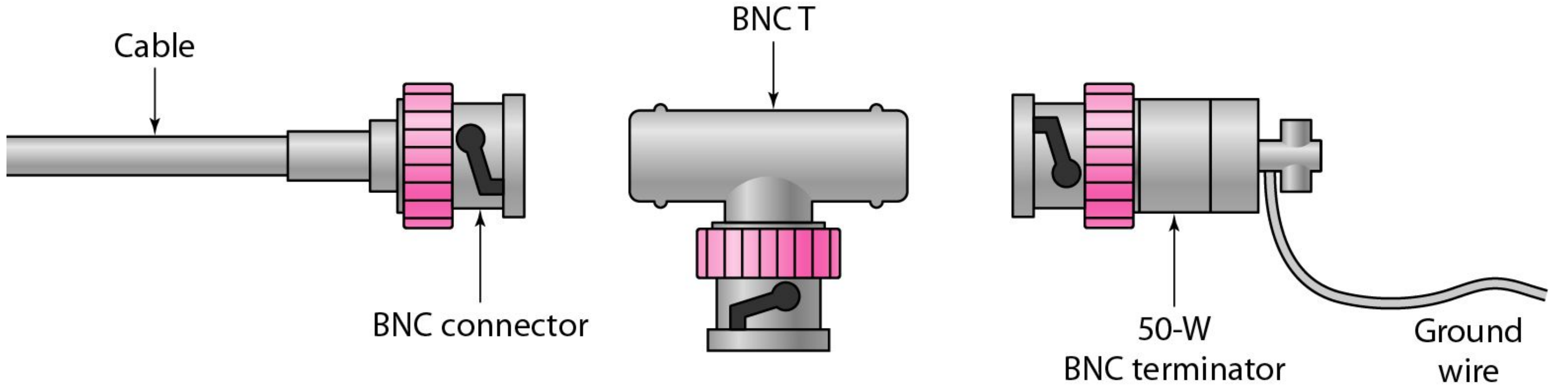
In-building networks at our university are almost all twisted pair

Coaxial cable

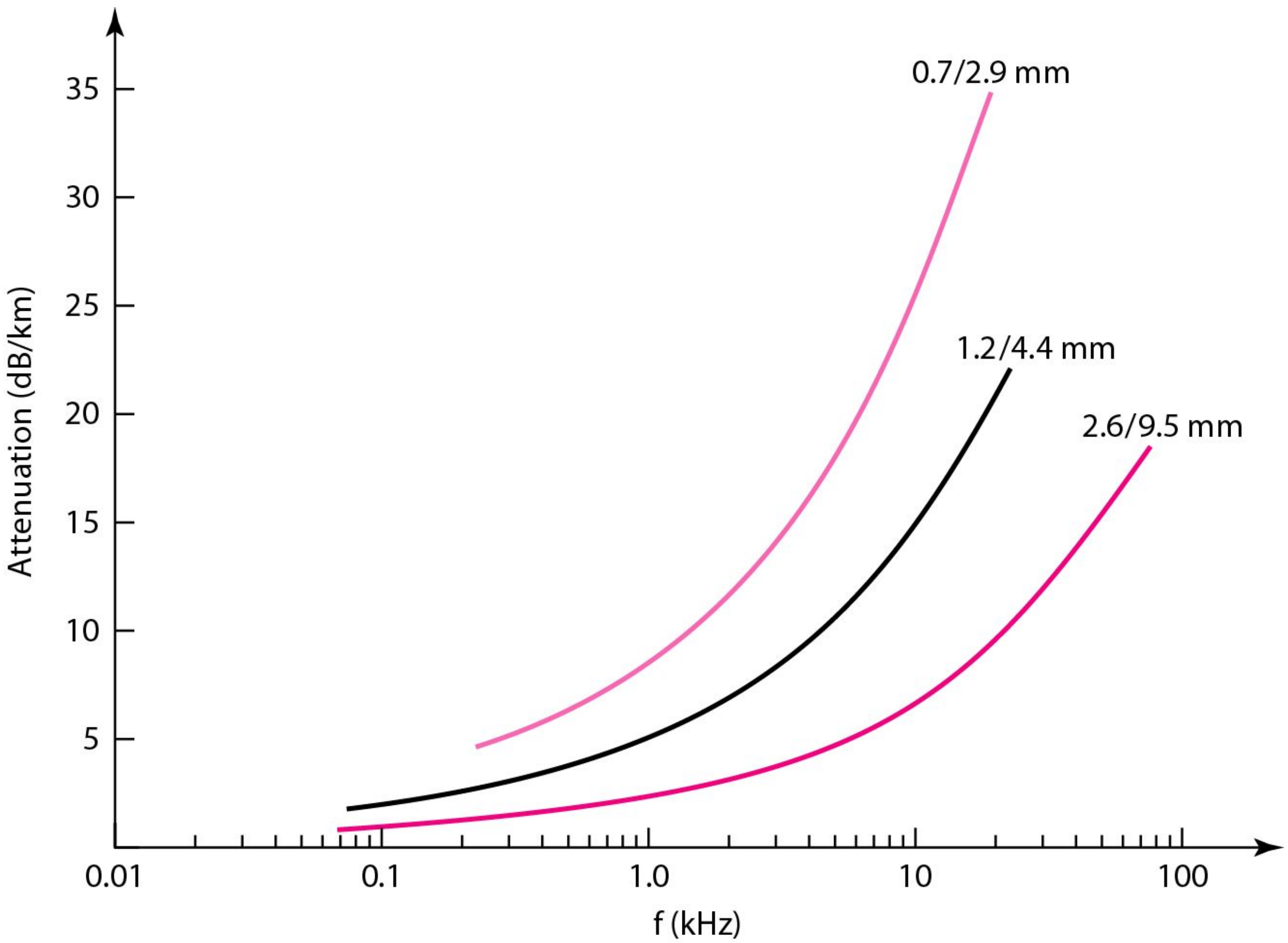


Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

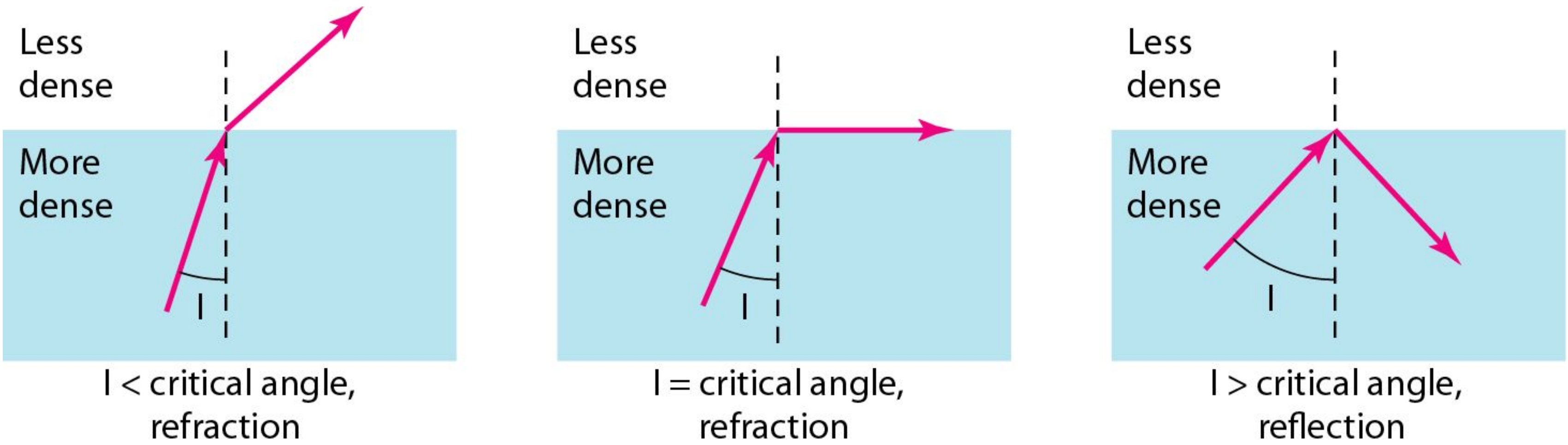
BNC connectors



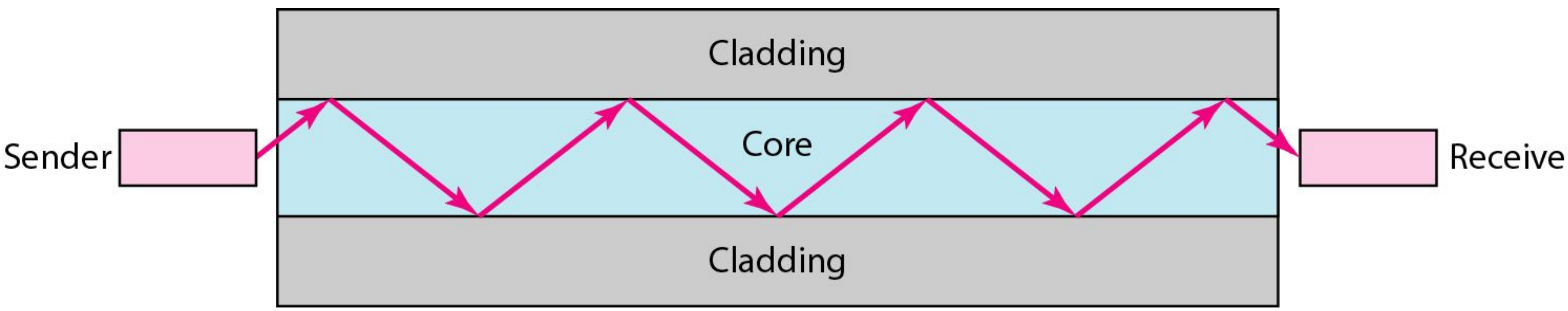
Performance Coaxial Cable



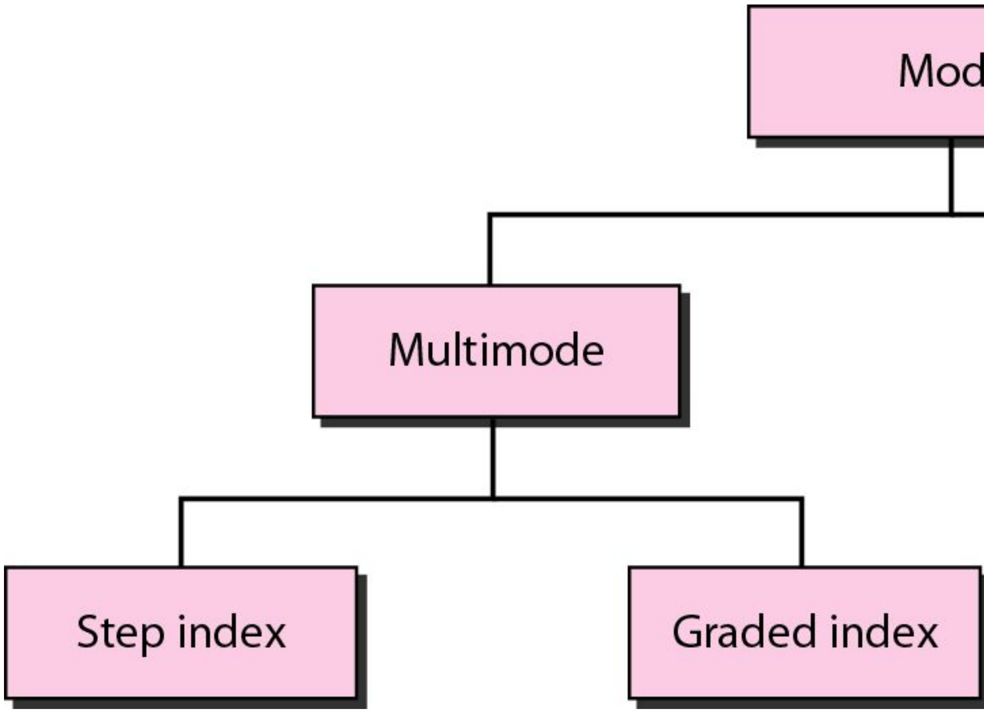
Bending of light ray



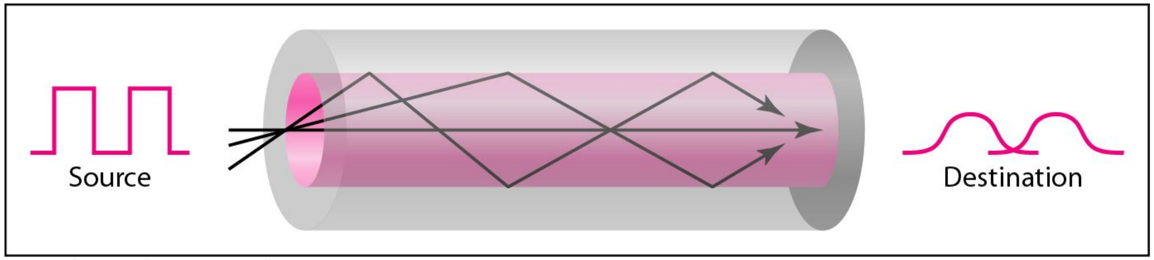
Optical fiber



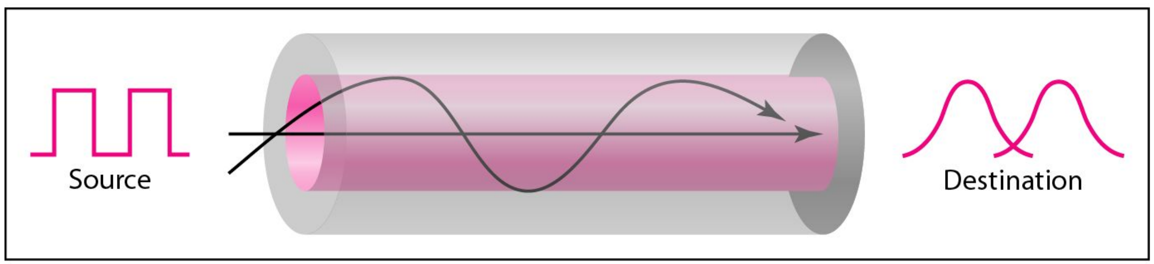
Propagation Modes



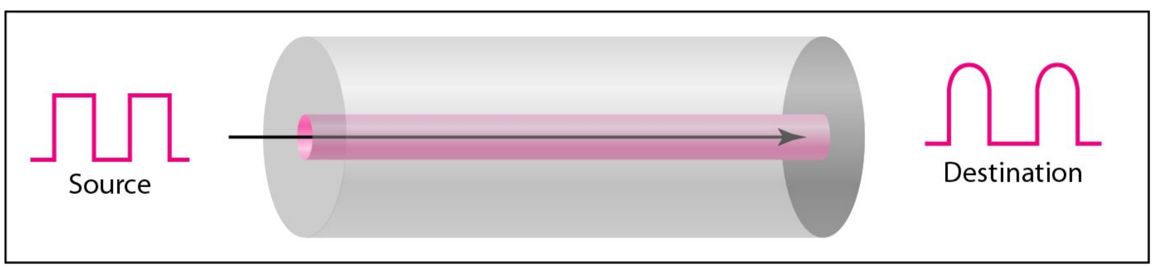
Modes



a. Multimode, step index



b. Multimode, graded index

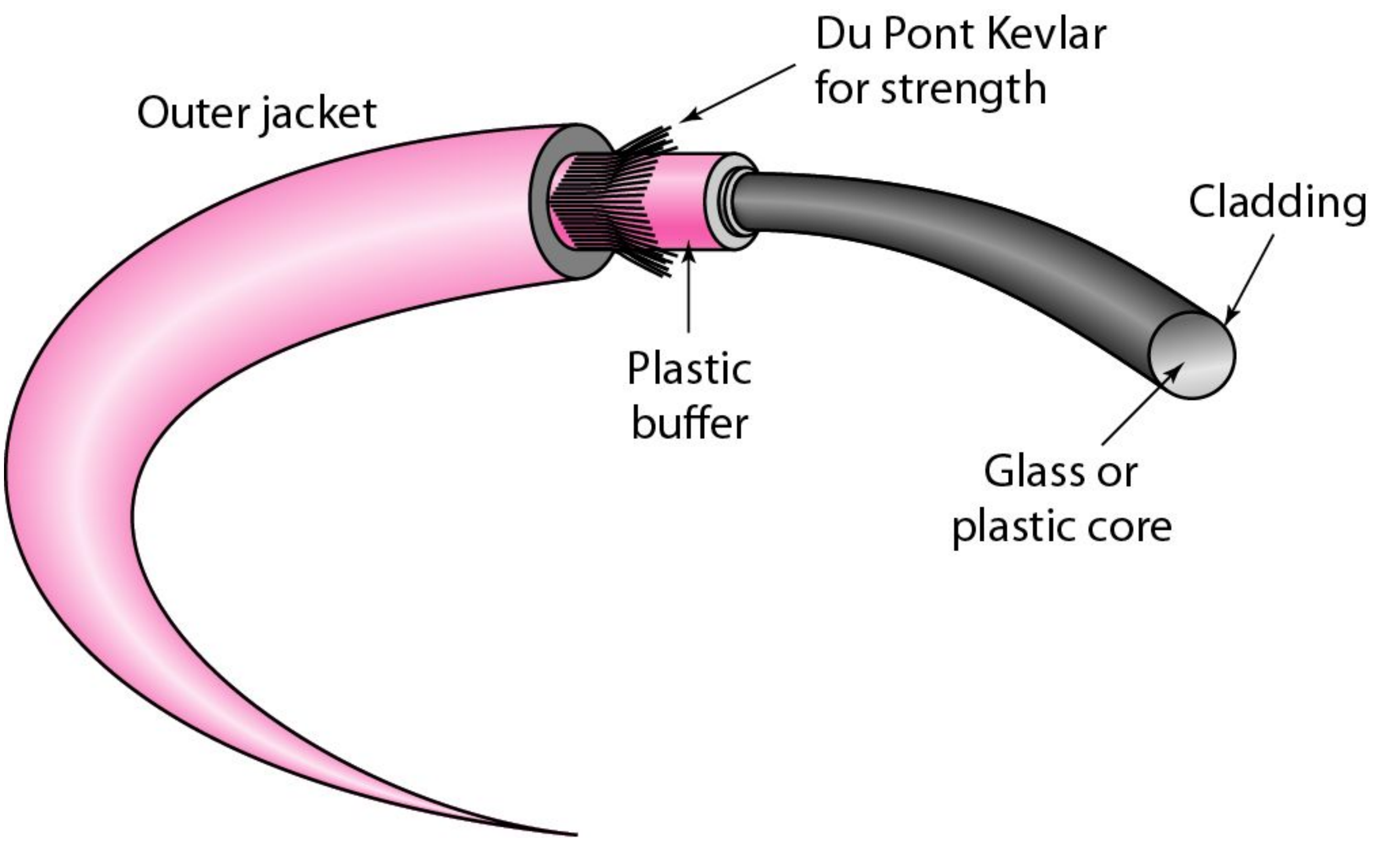


c. Single mode

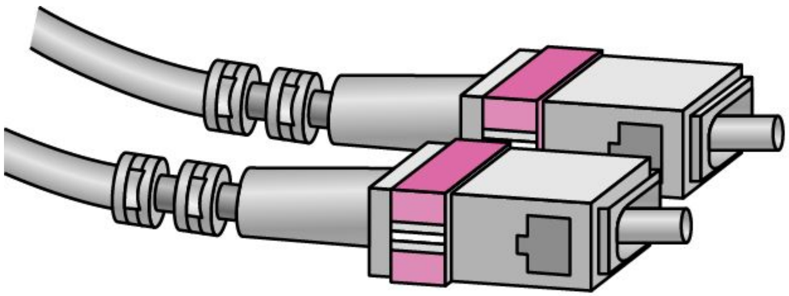
Fiber types

Type	*Core (μm)*	*Cladding (μm)*	*Mode*
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

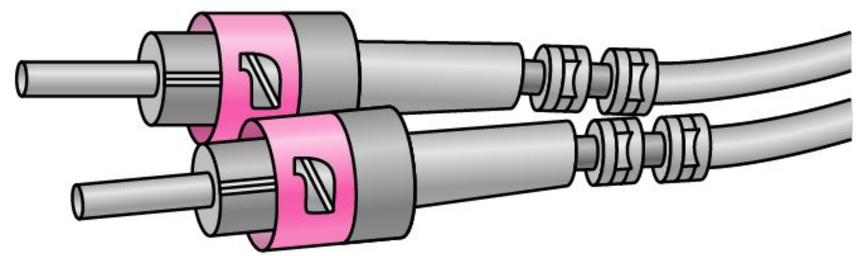
Fiber construction



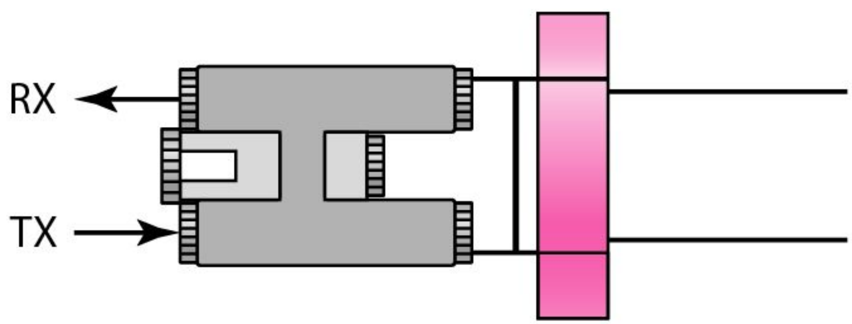
Fiber-optic Cable Connectors



SC connector

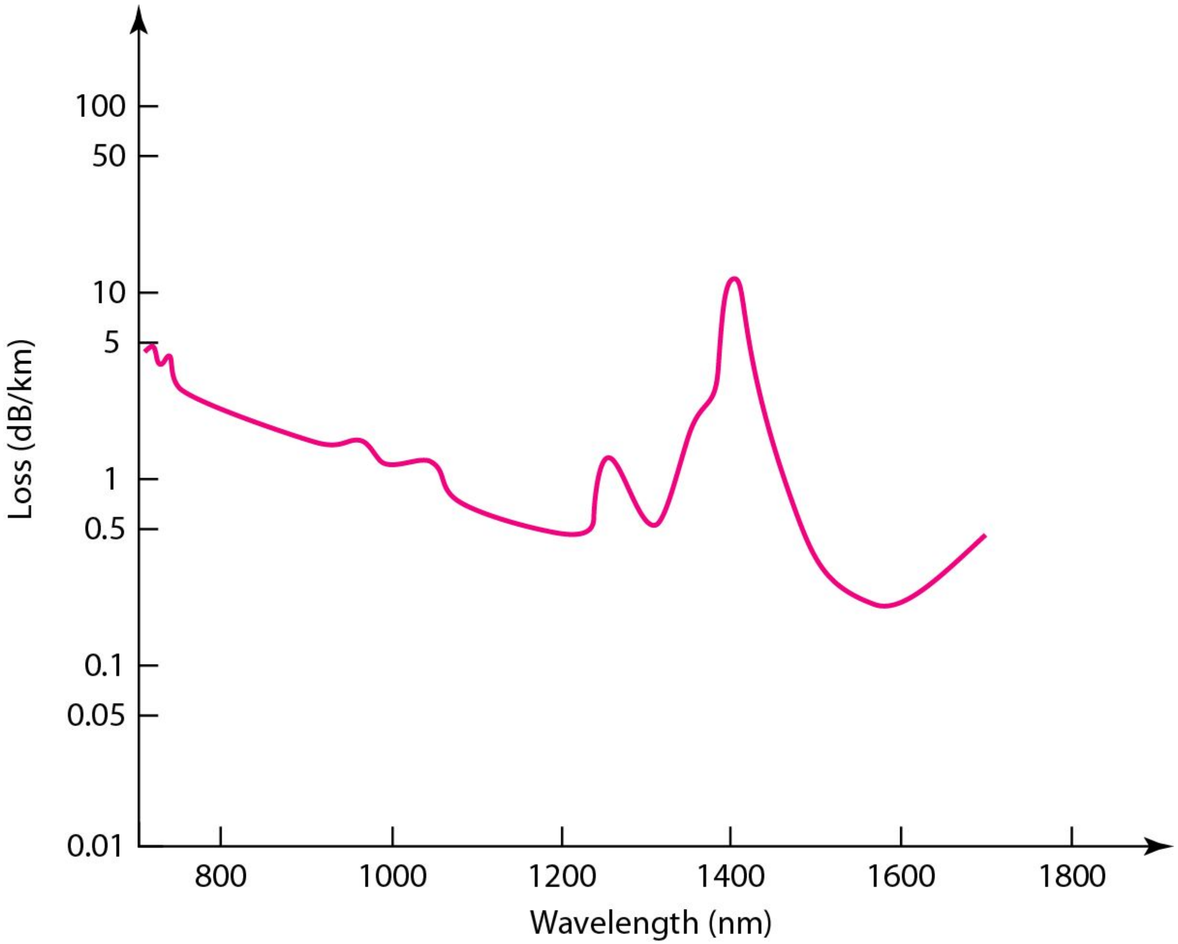


ST connector



MT-RJ connector

Performance Optical Fiber



Optical Fiber (Pros & Cons)

Pros:

- Low attenuation
- Large bandwidth

Cons:

- Relatively “new” technology
- “Expensive”

Comparing optical fiber to UTP

Pros:

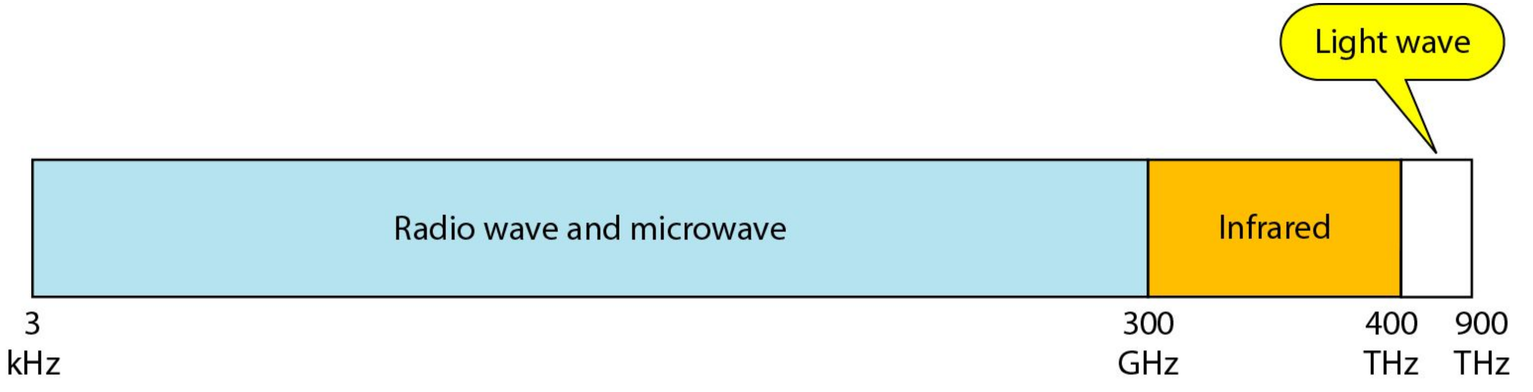
- Immune to electro-magnetic interference
  - no crosstalk
- Reduced need for error detection and correction
- Enables longer link distances
- Attenuation unaffected by transmission rate
- Easier network upgrade
- Can combine different services: telephony, TV, internet…

Cons:

- Optical components have higher cost
- Expensive deploying

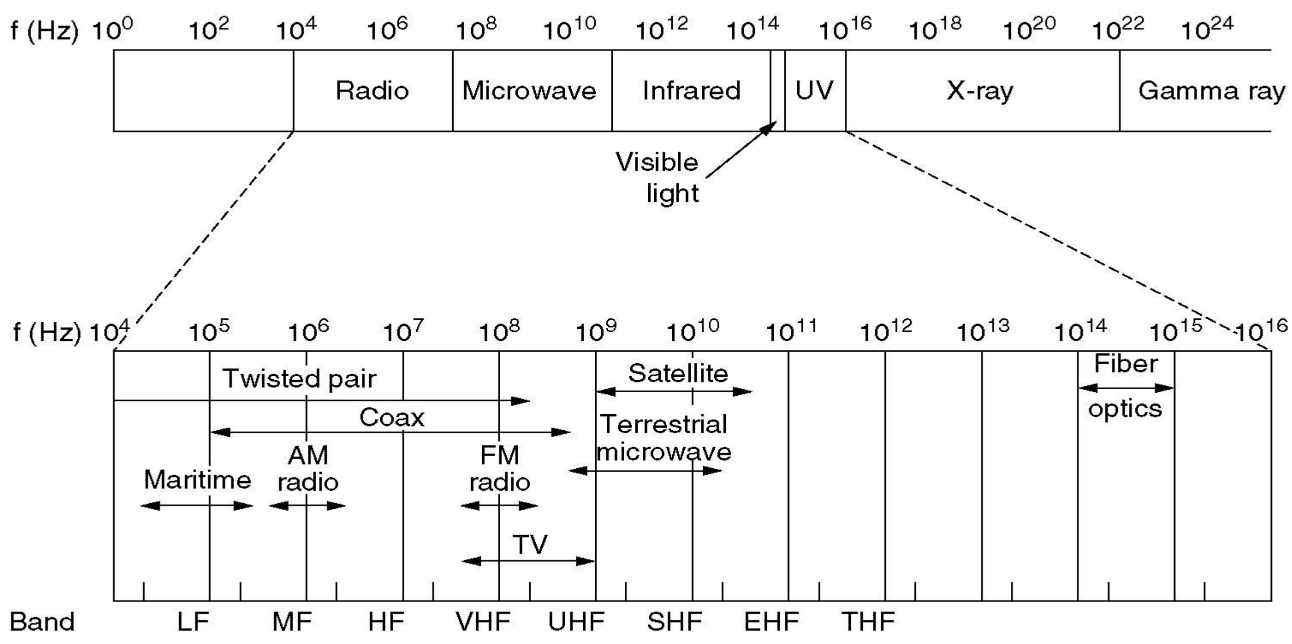
Unguided Media: Wireless

- Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.

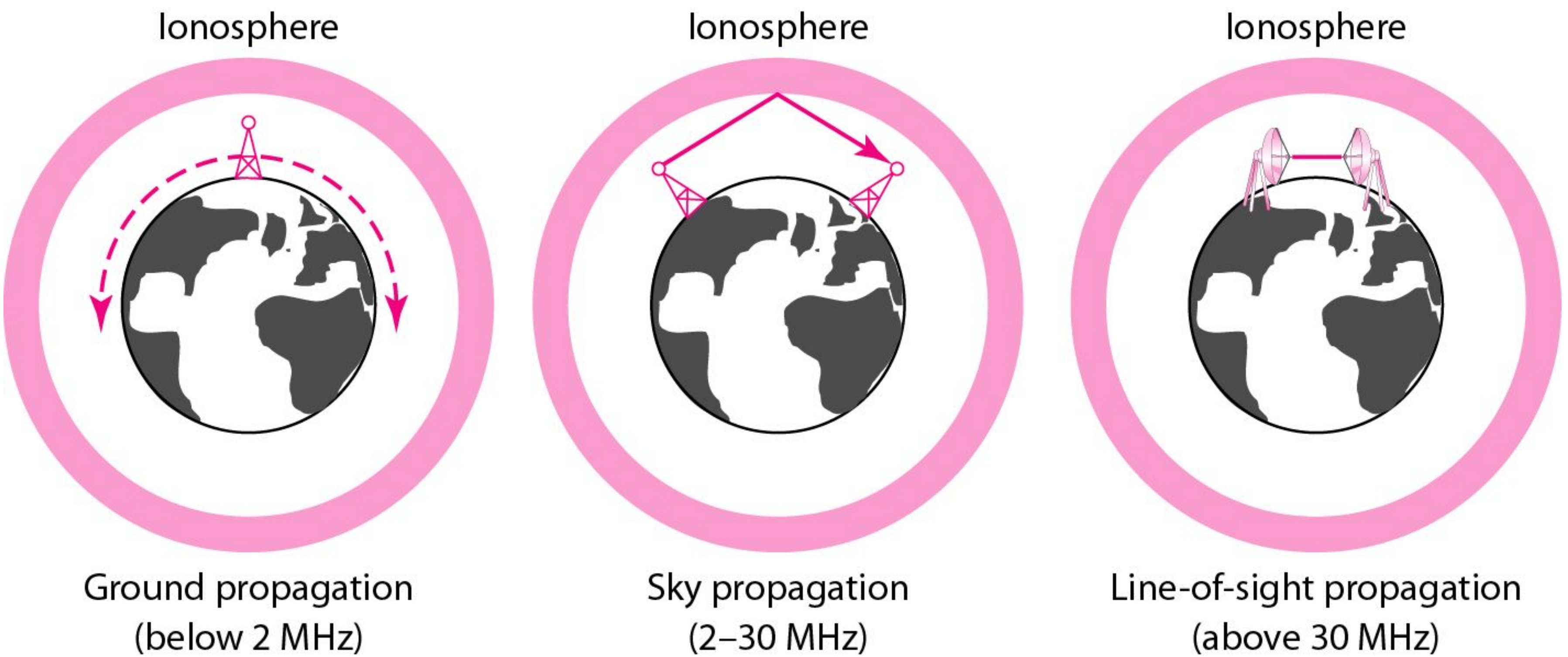


Wireless

- Modern wireless digital communication began in the Hawaiian Islands
- What is “the best” frequency to use for communication?



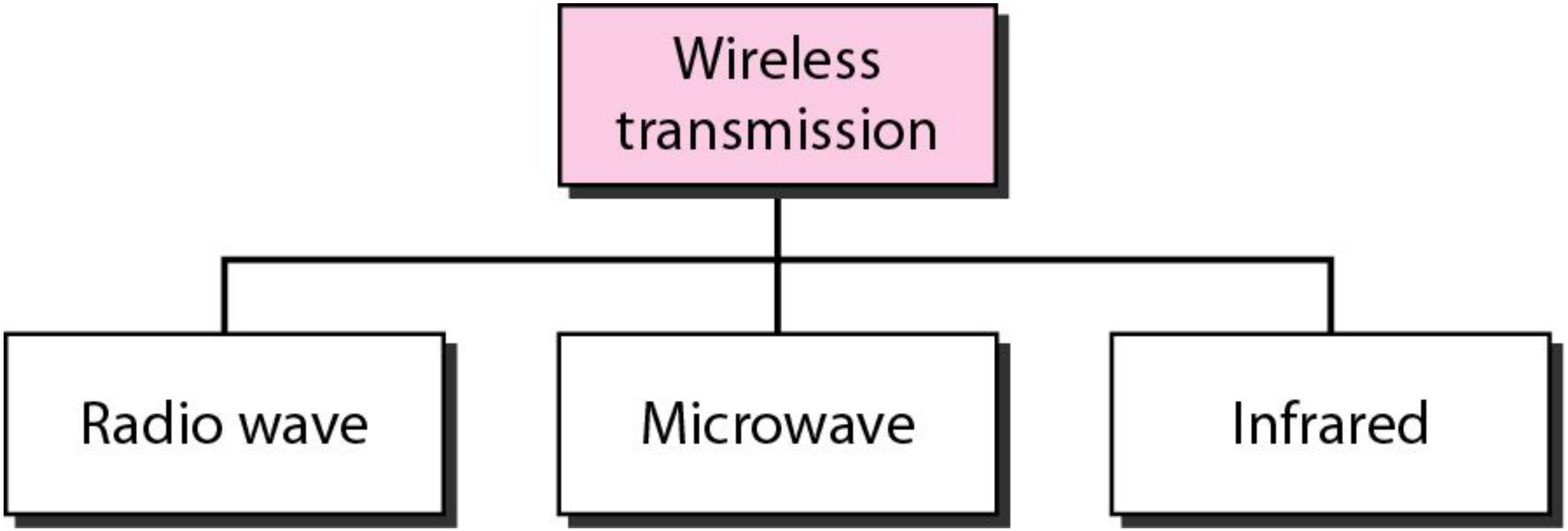
Propagation Methods



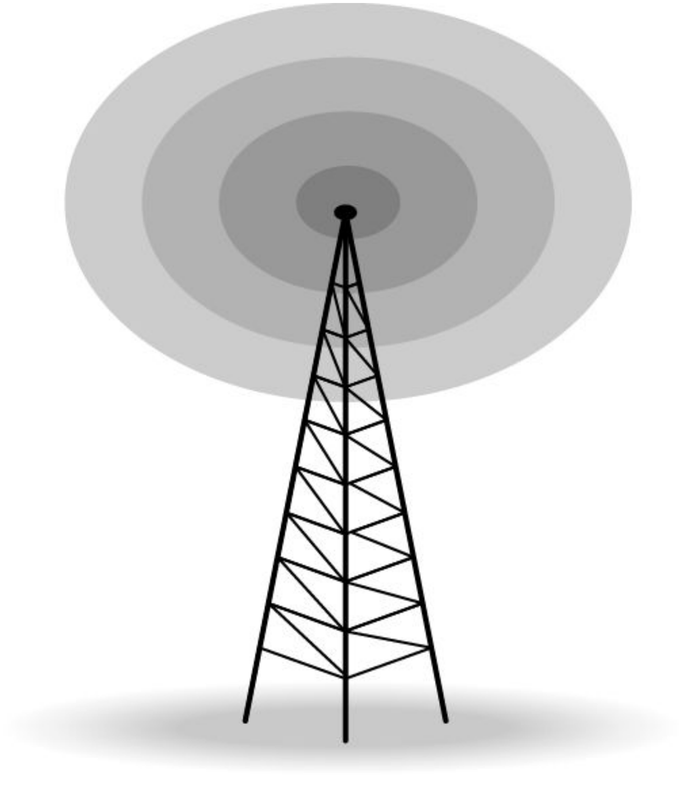
Bands

Band	*Range*	*Propagation*	*Application*
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30–300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

Wireless Transmission Waves



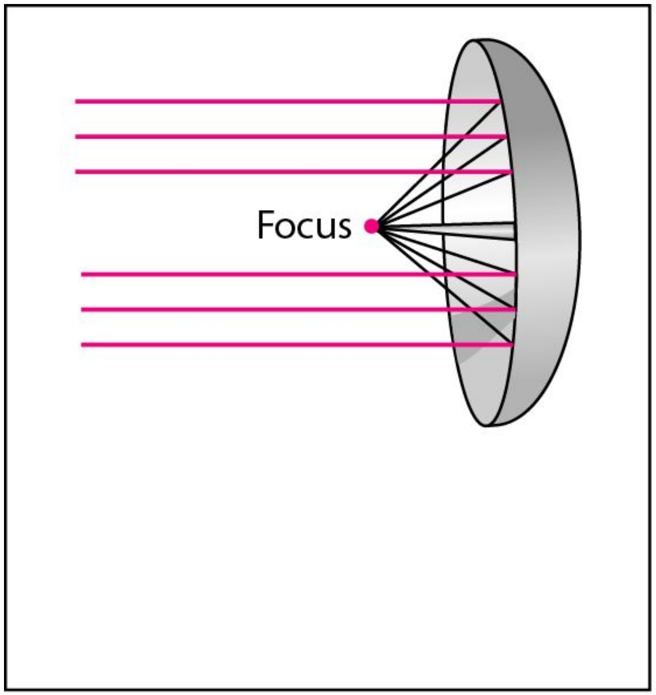
Omni directional Antenna



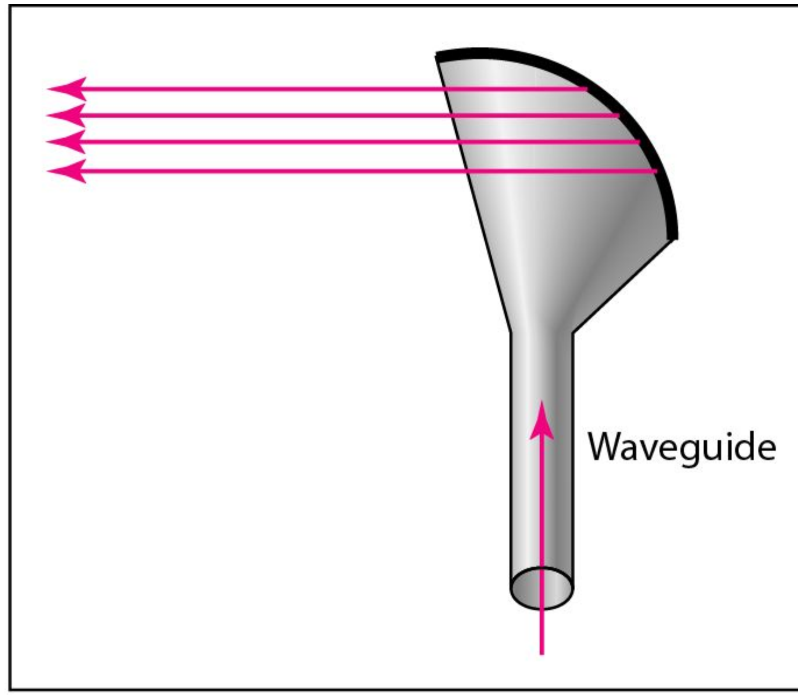
Note

Radio waves are used for multicast communications, such as radio and television, and paging systems.

Unidirectional Antennas



a. Dish antenna



b. Horn antenna

Note

Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

Note

Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

Readings

- Chapter 7 (B.A Forouzan)
  - Section 7.1, 7.2